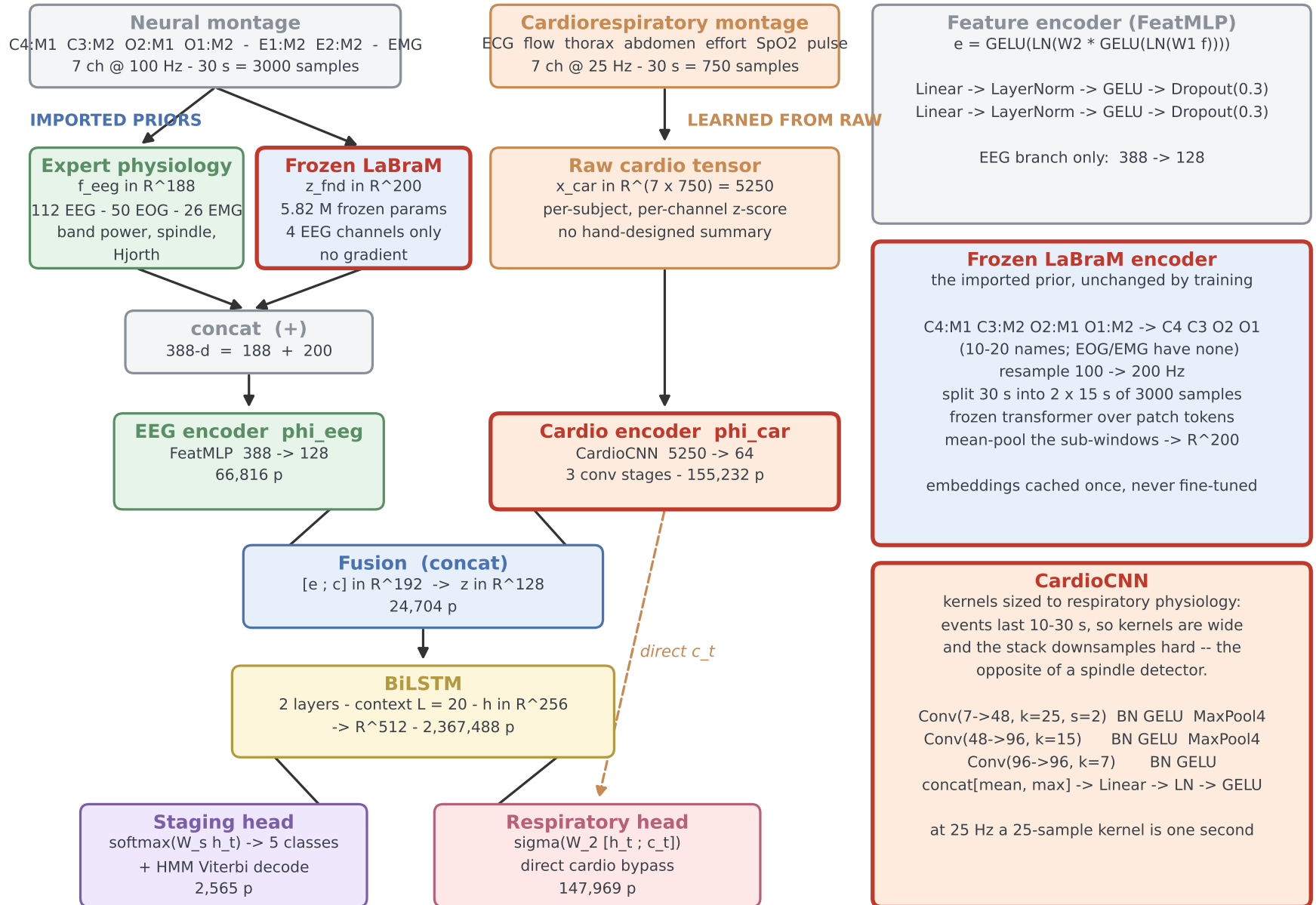


MM-Net v4: dual-prior neural branch, learned cardiorespiratory encoder

one 30 s polysomnogram epoch -> sleep stage + respiratory event

red outline = new since submission



Joint objective: $L = \text{CE_sqrt-w}(y_{\text{stg}}) + \lambda * \text{BCE_pw}(y_{\text{apn}})$, $\lambda = 1$

2,764,774 trainable parameters (773,254 in the submitted model) + 5,819,936 frozen

submitted 0.7275 acc / 0.6536 mF1 / 0.7070 AUC -> v4 0.7394 / 0.6701 / 0.7818